HEALTHIER, LONGER, BETTER LIVES

Medical Receipt OCR Benchmark

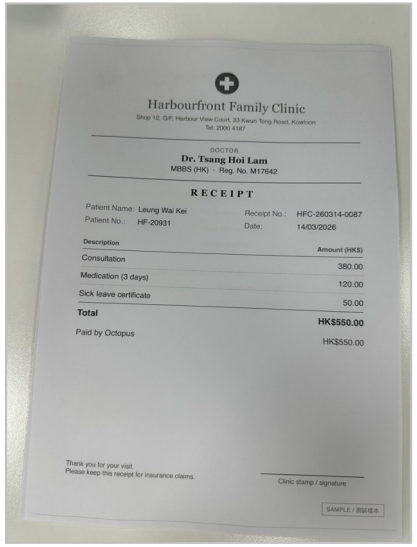
Can Azure OpenAI replace hand-written extraction rules?

9 receipts · 6 fields · manual rules vs AI extraction

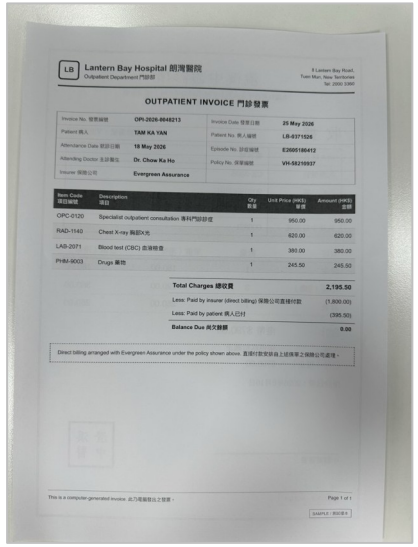
AIA confidential and proprietary information. Not for distribution.

What We Tested

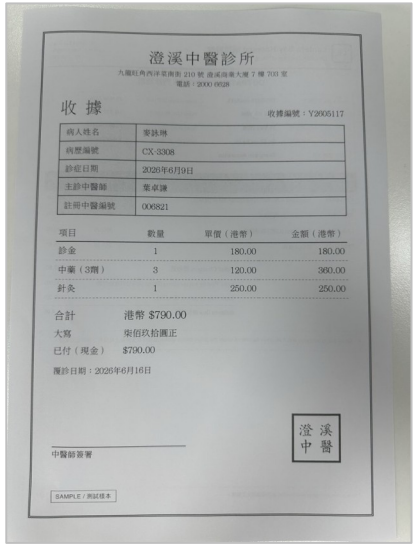
9 receipts with different layouts and languages. Each approach reads the text, then finds 6 fields.



English



Bilingual



Chinese

Approach	1. Read text	2. Find fields
PaddleOCR Light	Open-source OCR	Rules
Azure DI	Azure OCR	Rules
Azure DI + AOAI	Azure OCR	Azure OpenAI

Fields: doctor, registration no., patient, receipt no., service date, total.
51 fields scored.

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]

Results: One Fair Accuracy Figure

AI extraction reached 88.2% with a prompt written before these receipts: the only untuned result.

Approach	Accuracy (9 receipts)	Time / receipt	How to read it
PaddleOCR Light + rules	Not shown	13.9 s	Rules tuned by hand on these receipts
Azure DI + rules	Not shown	3.8 s	Rules tuned by hand on these receipts
Azure DI + AOAI, original prompt	**88.2%** (45/51)	6.2 s	**Prompt written before these receipts**
Azure DI + AOAI, prompt v2	96.1%* (49/51)	6.4 s	Development result: prompt tuned on these receipts

Unseen receipts (Version 1): tuning did not carry over. The improved rules scored the same as the original rules on 4 unseen receipts, and prompt v2 the same as the original prompt on 6 receipts. Too few to rank approaches.

*Rules: no accuracy shown (original rules had no Chinese labels; improved rules were tuned on these receipts). * Not independently validated.*

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]

AOAI Replaces the Manual Rule Layer

Both pipelines read the text the same way. Only the step that finds the fields changes.

Rule-based
PaddleOCR Light + rules
tuned by hand for these layouts

Read text (OCR)
PaddleOCR Light

Extraction rules
written and updated by hand
(69 → 225 label phrases)

6 fields
doctor, patient, total, …

replaced by

AI extraction
Azure DI + AOAI
88.2%, original prompt

Read text (OCR)
Azure DI

Azure OpenAI
one prompt, written before these receipts

6 fields
doctor, patient, total, …

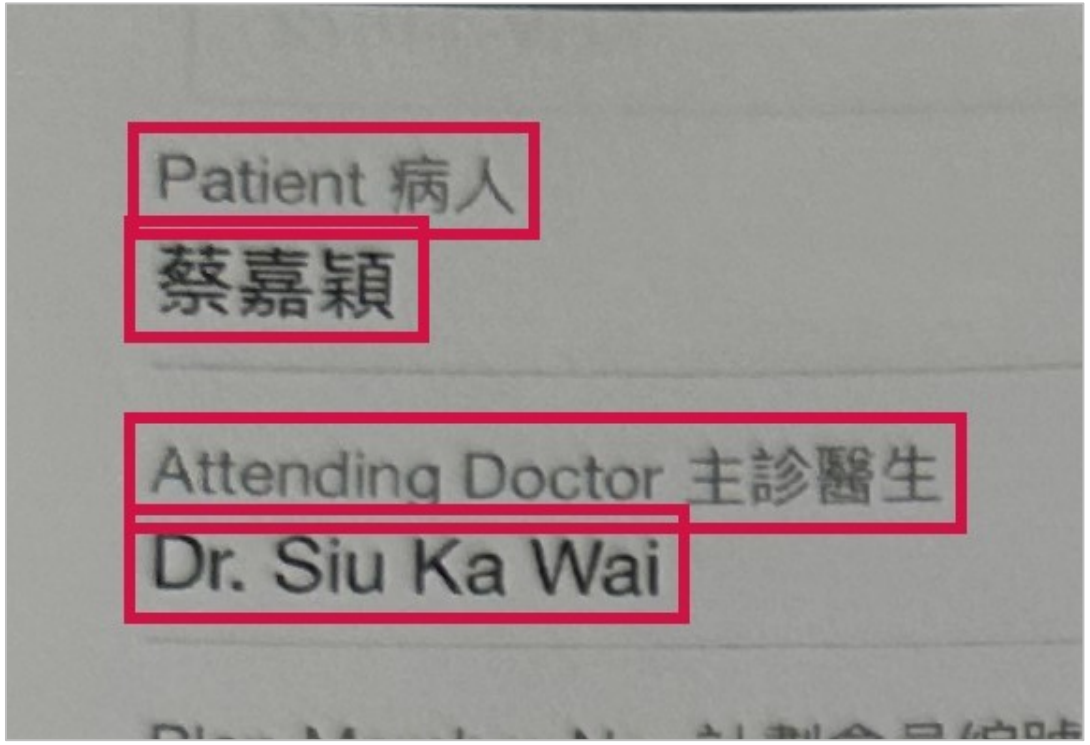
AOAI replaces the manual rule layer. It still makes mistakes (6 of 51 fields wrong with the original prompt) and has not yet been tested on a large set of unseen receipts.

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]

Example 1: The OCR Was Right, the Rules Were Wrong

Both OCR engines read the names. The original rules took the label 病人 as the patient's name.



Field	Correct	Original rules	Improved rules	DI + AOAI
Patient	蔡嘉穎	✗ 病人	✓ 蔡嘉穎	✓ 蔡嘉穎
Doctor	Dr. Siu Ka Wai	✗ not found	✓ Dr. Siu Ka Wai	✓ Dr. Siu Ka Wai

- Original rules: took the Chinese label 病人 ("patient") as the patient's name.
- Improved rules: fixed by adding bilingual labels by hand.
- AOAI: correct with its original prompt; no change needed.

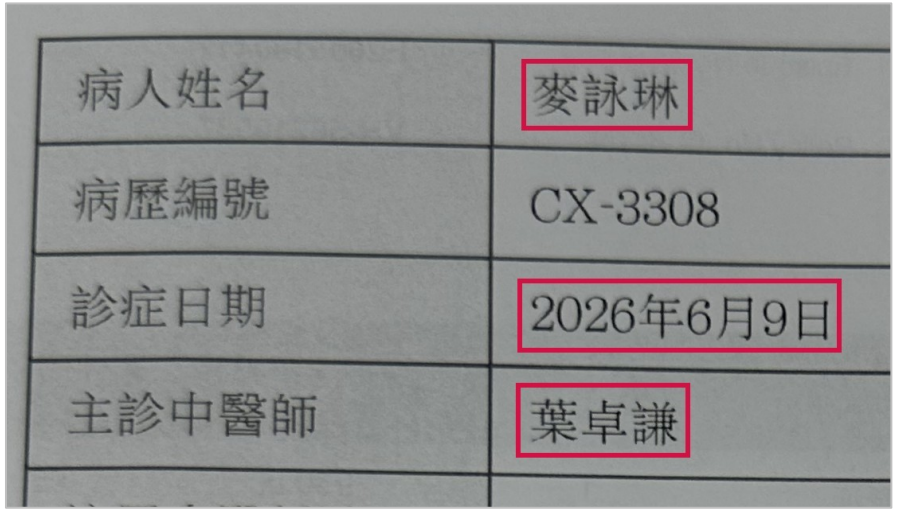
Bilingual medical centre receipt. Rules shown on PaddleOCR Light text; Azure DI with the original rules made the same patient error. DI + AOAI = original prompt.

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]

Example 2: Every New Layout Meant Manual Rule Work

Each label wording was added by hand, one change broke a receipt, and the tuning did not carry over.



Chinese medicine receipt: original rules 0/6 fields; after Chinese labels were added by hand 6/6; DI + AOAI 6/6 (original prompt).

69 → 225

label phrases in the rules, added by hand after reviewing the errors

369303 → 380048

vet invoice receipt number: correct before the changes, wrong after (a regression)

No gain

on 4 unseen receipts: the tuned rules scored the same as the original rules

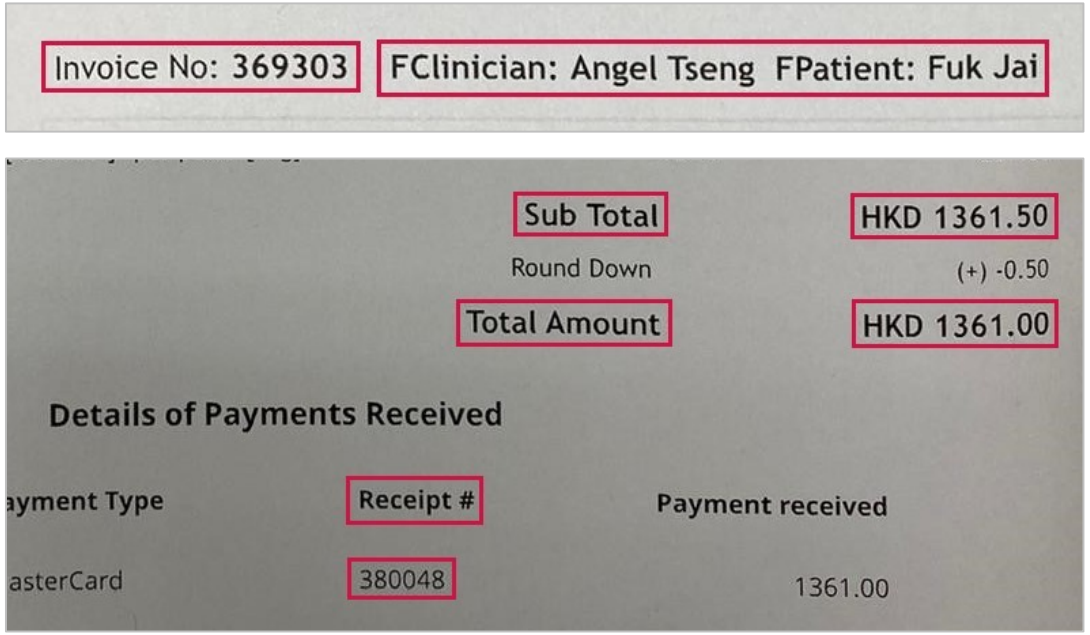
With its original prompt, DI + AOAI needed none of these changes for these receipts. Whether that holds for new layouts still needs to be tested.

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]

Example 3: AOAI Still Makes Mistakes

On the vet invoice, the improved rules and AOAI each got a different field wrong.



Field	Correct	Original rules	Improved rules	DI + AOAI
Receipt no.	369303	✓ 369303	✗ 380048	✓ 369303
Patient	Fuk Jai	✗ not found	✗ not found	✓ Fuk Jai
Total	1361	✓ 1361.0	✓ 1361.0	✗ 1361.5

- Improved rules: picked the payment receipt number.
- Rules: the OCR read "FPatient:", so the patient label was missed.
- AOAI (original prompt): picked the sub total. Prompt v2 fixes it, but was tuned on this receipt.

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]

Development Effort and Maintenance

The difference is the manual work to handle each layout, and to keep it working.

	PaddleOCR Light + rules	Azure DI + AOAI
Accuracy (9 receipts)	Not a fair measure: rules tuned on these receipts	88.2% original prompt; 96.1% prompt v2 (development)
Effort to get there	Rules rewritten by hand after reviewing errors on these receipts	None for the original prompt; one revision for prompt v2
Extraction logic	69 → 225 label phrases to maintain	One prompt and a few general checks
Unseen receipts	Tuning gave no gain (4 receipts)	Prompt v2 gave no gain (6 receipts)
Errors seen	A rule change broke a receipt that was already correct	6 of 51 fields wrong (original prompt), e.g. a total
Time and running cost	13.9 s, own compute, no API fee	6.2 s, Azure fees per receipt (to confirm)

The main question is not API price but who maintains the extraction logic as layouts change. Lower maintenance for AOAI is implied by this benchmark, not yet proven.

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]

Manual Rules vs. AI Extraction: Pros and Cons

"Measured" = this benchmark; the rest is expected.

Manual extraction rules (PaddleOCR Light)

New formats need manual rule updates, which can break working cases and add maintenance and testing.

Pros

- Full control; transparent and predictable on formats the rules know.
- No API fees: open-source OCR on our own compute.
- **Measured:** the OCR read the values; the misses were missing labels.

Cons

- Each new format or language needs rule updates by hand.
- **Measured:** a rule change broke a receipt that was already correct.
- **Measured:** 69 → 225 label phrases; no gain on unseen receipts.

AI extraction (Azure DI + AOAI)

Less rule maintenance expected: accuracy depends mainly on the AI model, OCR output and prompt.

Pros

- **Measured:** 88.2% with a prompt written before these receipts.
- Reads varied formats using existing model capabilities.
- **Measured:** faster, 6.2 s vs 13.9 s per receipt.

Cons

- **Measured:** still wrong on 6 of 51 fields, e.g. a total.
- Less direct control: results depend on model, OCR, prompt.
- API fees; needs validation, monitoring and prompt upkeep.

Neither is maintenance free. The difference is where the effort sits: rules for every layout, or one prompt plus validation.

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]

Conclusion and Next Steps

The real trade-off: maintain extraction rules ourselves, or let AI replace that layer.

- Azure DI + AOAI reached 88.2% with a prompt written before these receipts. A tuned prompt reached 96.1% on the same receipts: a development result, not validated.
- The rules worked only after hand-tuning for these layouts, and the tuning did not help on unseen receipts.
- **So the choice is whether to keep writing and maintaining extraction rules, or to use AI for that layer.**

1. Freeze both

Lock the improved rules and AOAI prompt v2.

2. Test on unseen receipts

30+ real, anonymised receipts. Measure accuracy and count the rule changes needed.

3. Then compare cost

Compute, Document Intelligence pages and gpt-5-mini tokens.

Limitation: 9 receipts, mostly printed mock receipts; the rules and prompt v2 were tuned on them. AOAI still makes mistakes and has not been tested on a large unseen set.

AIA confidential and proprietary information. Not for distribution.

[AIA – PUBLIC]